



TECHFEST PROTOTYPE • IoT WATER MANAGEMENT

JALRAKSHAK

Smart IoT System to Monitor Water Distribution, Detect Leakages & Prevent Theft

Real-time flow monitoring, automated leak detection, and instant alerts to protect every drop of water.



CATEGORY

IoT Smart Water Management



PURPOSE

TechFest Prototype Demo



CENTRE

DigiSkill Khar Road

Save Water. Save Life. Smart Monitoring. Better Future.

Water Loss Is a Growing Crisis



Silent Leakages

Pipeline leaks in distribution networks often go undetected for long periods, wasting huge volumes of water.



Water Theft

Unauthorized tapping and illegal connections divert water away from intended supply, causing shortages.



Delayed Detection

Without sensors, leaks and theft are usually discovered only after major loss or complaints.



Irregular Supply

Undetected loss and theft disrupt planning, leading to inconsistent and unfair water distribution.

JalRakshak ensures efficient monitoring and management of water distribution — detecting leaks and theft in real time.

Key Features of JalRakshak



Real-Time Flow Monitoring

Continuously tracks water flow at both inlet and outlet points.



Leakage Detection

Identifies leaks by comparing inlet and outlet flow readings.



Water Theft Detection

Flags abnormal flow patterns that indicate unauthorized tapping.



Automatic Valve Control

Shuts off the control valve automatically when danger is detected.



Instant Alerts

Sends immediate LED, buzzer, and notification-based alerts.



IoT Remote Monitoring

Enables dashboard and mobile-based monitoring from anywhere.

Two Scenarios: Safe vs. Danger

Twin flow sensors at the inlet and outlet continuously compare readings to detect leakage or theft.

SCENE 1: NORMAL FLOW

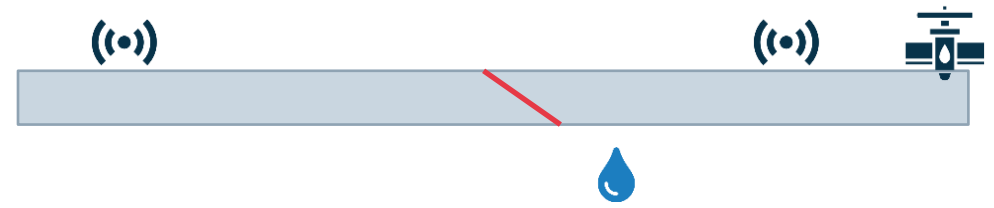


STATUS: SAFE

- ✓ Flow Normal, No Leakage
- ✓ No Theft Detected
- ✓ Green LED ON
- ✗ Buzzer OFF

Control Valve: OPEN

SCENE 2: LEAKAGE / THEFT



STATUS: DANGER

- ✗ Abnormal Flow Detected
- ✗ Possible Leakage / Theft
- ✗ Red LED ON
- ✓ Buzzer ON — Alert Sent

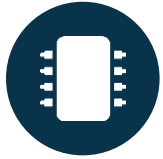
Control Valve: CLOSED

System Architecture



Components Used

01



ESP8266 NodeMCU

Main microcontroller & Wi-Fi module

02



YF-S201 Flow Sensor (x2)

Measures inlet & outlet water flow

03



Motorized / Solenoid Valve

Controls water flow automatically

04



Green LED

Indicates safe condition

05



Red LED

Indicates danger condition

06



Buzzer

Sounds an audible alert

07



Power Supply (5V)

Powers the complete circuit

08



Connecting Wires

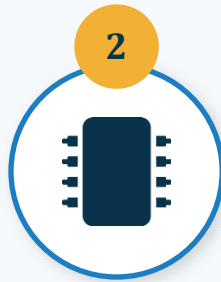
Circuit interconnections

How It Works



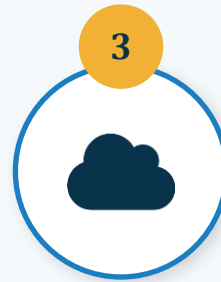
Flow Sensors

measure water flow at inlet and outlet



ESP8266

reads data and sends it to the cloud



Cloud Analysis

detects abnormal flow difference



Alert Triggered

LED, buzzer & notifications activate



Authorities

take action based on the alert

Status Indication (LED & Buzzer)

CONDITION	GREEN LED	RED LED	BUZZER
Safe (Normal Flow)	ON	OFF	OFF
Danger (Leakage / Theft)	OFF	ON	ON



Why It Matters

Simple, unmistakable LED and buzzer cues let field staff and citizens instantly recognize a leak or theft event — no technical training required.

Project Guides & Participants



Project Guides

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Centre: DigiSkill Khar Road

Category: IoT Based Smart Water Management System

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Thank You

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